NORTHWIND INTERACTIVE SALES DASHBOARD

Using Microsoft Power BI

PROJECT REPORT

Submitted By:

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Tool Used: Microsoft Power BI Desktop

1. Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps me organize data and present useful information through interactive reports, charts, tables and KPI cards.

For this project, I created a Northwind Interactive Sales Dashboard using Microsoft Power BI. I used order, customer, product, category, employee, country, quantity and freight information to study sales activity and customer distribution.

The dashboard helps me monitor order activity by month and employee, compare sales across product categories, review customer counts by country and examine product-level sales and quantities. I also added filters and a category tooltip page to make the report interactive.

2. Objectives of the Project

The main objectives of my project are:

- To summarize order, customer, quantity and freight information using KPI cards.
- To analyse the monthly pattern of total orders.
- To compare the number of orders handled by employees.
- To compare total sales and quantity sold across products and categories.
- To analyse the distribution of customers across countries.
- To create an interactive dashboard using year, country and category filters.
- To display additional category-level information through a tooltip page.

3. Dataset Description

I used Northwind sales data containing information about orders, customers, products, categories, employees, countries and freight. The report uses product and category names for sales comparisons, employee names for order comparisons, and country names for customer analysis.

The main fields and measures used in my dashboard are:

Field / Measure	Description
Order Count / Total Orders	Counts the orders represented in the report.
Number of Customers	Shows the number of customers included in the analysis.
Total Quantity Sold	Displays the total quantity of products sold.
Total Freight	Displays the freight amount associated with the orders.
Average Freight	Shows the average freight value.
Month	Used to analyse total orders across the months.
EmployeeName	Identifies employees used in the order-count comparison.
ProductName / categoryName	Identifies products and product categories used for sales and quantity analysis.
Country	Used to compare the number of customers by country.
Year	Used to filter the dashboard by year.

4. Data Preparation

I arranged the available information so that I could analyse it by order month, employee, product, category and country. I used the order and customer fields for the summary cards, product and category fields for the sales table and category chart, and country for the customer distribution chart.

I also used the numeric fields for order count, quantity, sales and freight summaries. Date and year fields support the time-based chart and year filter. Checking the data types and ensuring that numeric fields are treated as numbers is important for accurate aggregation in Power BI.

5. DAX Measures and KPI Cards

I used summary measures to display the main order and freight information in the KPI section. The cards show:

- **Total Orders:** 809.
- **Number of Customers:** 91.
- **Total Quantity Sold:** approximately 51K.
- **Total Freight:** approximately 63.96K.
- **Average Freight:** 79.05.

The product-level table displays total sales of 1,265,793.04 and total quantity of 51,317. The tooltip page displays total sales as approximately 1.27M, quantity sold as 51K and total orders as 809. These are rounded display values.

6. Dashboard Components

I created different visualizations to analyse order activity, product sales and customer distribution.

6.1 Total Orders by Month

I used a line chart to show the total number of orders for each month from January to December. The order count is relatively high in March, at just over 100, and April is also high. The line drops sharply in May and reaches its lowest point around June, before recovering from July onward. December shows a further increase.

Purpose: To identify monthly changes in order activity and compare months with higher or lower order counts.

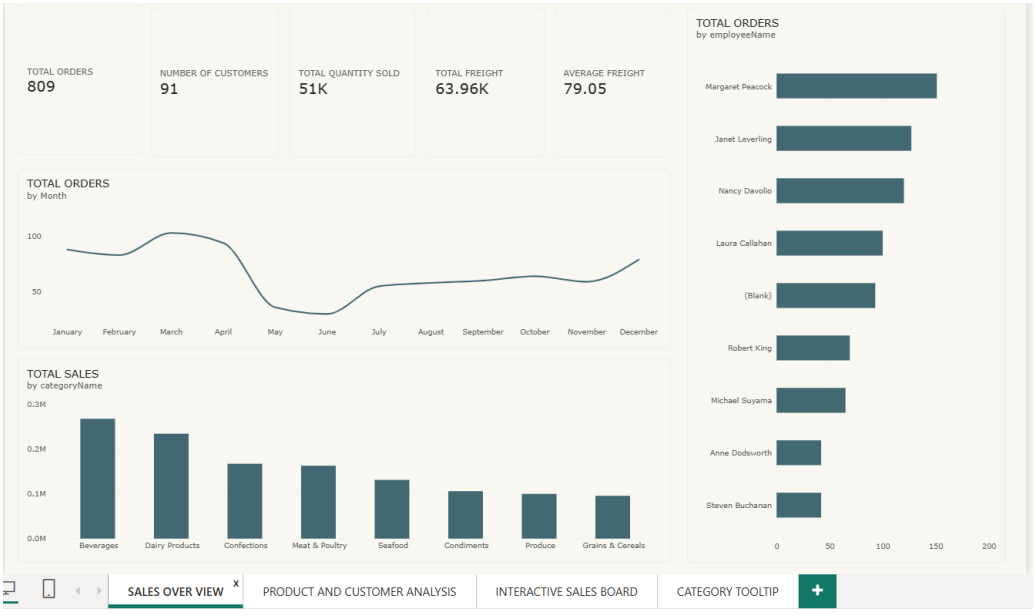


Figure 1: Monthly order trend and orders by employee.

6.2 Total Orders by Employee

I used a horizontal bar chart to compare total orders by employee. Margaret Peacock has the longest bar in the displayed chart, followed by Janet Leverling and Nancy Davolio. Laura Callahan and the blank employee category appear below them, while the remaining visible employees have shorter bars. This allows me to compare the order counts associated with each employee.

Purpose: To compare employee order activity and identify employees associated with higher or lower order counts.

6.3 Total Sales by Category

I used a column chart to compare total sales across the product categories. Beverages has the highest sales in the chart, followed by Dairy Products. Confections and Meat & Poultry are in the next group, while Seafood, Condiments, Produce and Grains & Cereals show lower sales values.

Purpose: To identify the categories contributing more or less to total sales.

6.4 Product-Level Sales and Quantity Table

I used a table to display ProductName, categoryName, Total Sales and Total Quantity Sold for individual products. The total row shows sales of 1,265,793.04 and quantity sold of 51,317. This table helps me inspect product-level performance rather than only viewing category totals.

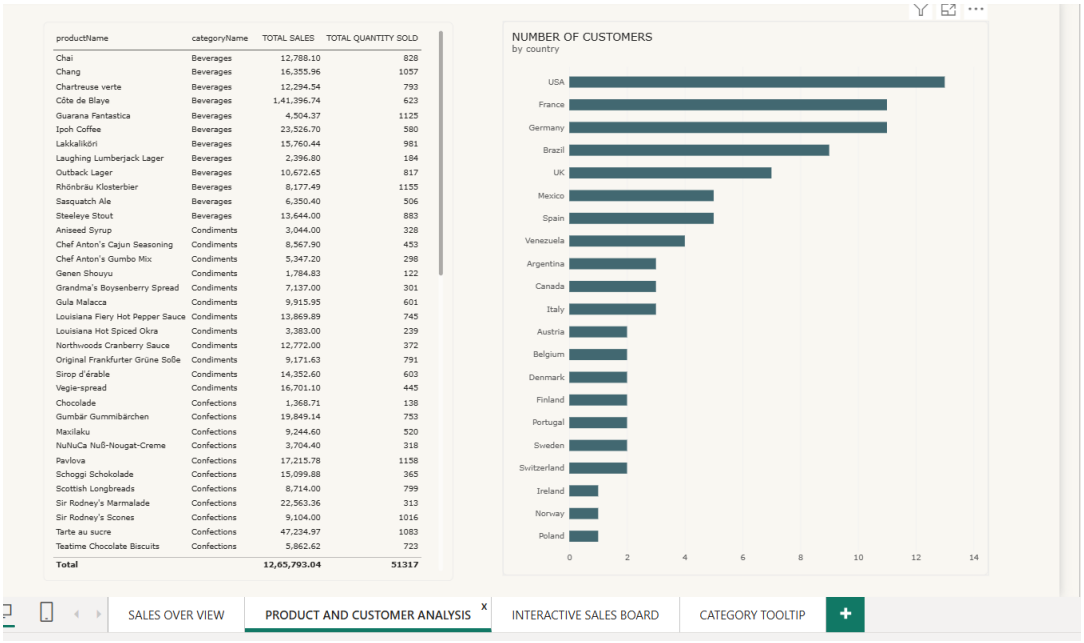


Figure 2: Product-level sales table and customer count by country.

6.5 Number of Customers by Country

I used a horizontal bar chart to compare the number of customers across countries. USA has the highest customer count in the displayed chart. France and Germany follow, with Brazil and the UK also showing relatively high counts. The remaining countries have smaller customer counts in comparison.

Purpose: To understand where the customers are located and compare customer distribution across countries.

7. Slicers Used

I added filters to make the main dashboard interactive:

- **Year:** I can select a year. The visible options include 2013 and 2014.
- **Country:** I can select a country to focus the dashboard on its sales and order information.
- **Category:** I can select a product category to analyse the related sales and orders.

These filters help me focus on a selected year, country or product category. The connected visuals can then be reviewed according to the selected filter values.

8. Key Findings

From my dashboard analysis, I identified the following findings:

- The dashboard shows 809 total orders and 91 customers.
- The total quantity sold is approximately 51K, while total freight is approximately 63.96K.
- Average freight is displayed as 79.05.
- Order activity is highest around March in the monthly trend and falls sharply during May and June.
- Margaret Peacock has the highest displayed order count among the listed employees.
- Beverages is the highest-sales category in the category chart, followed by Dairy Products.
- The product table reports total sales of 1,265,793.04 and total quantity sold of 51,317.
- USA has the highest displayed customer count by country.

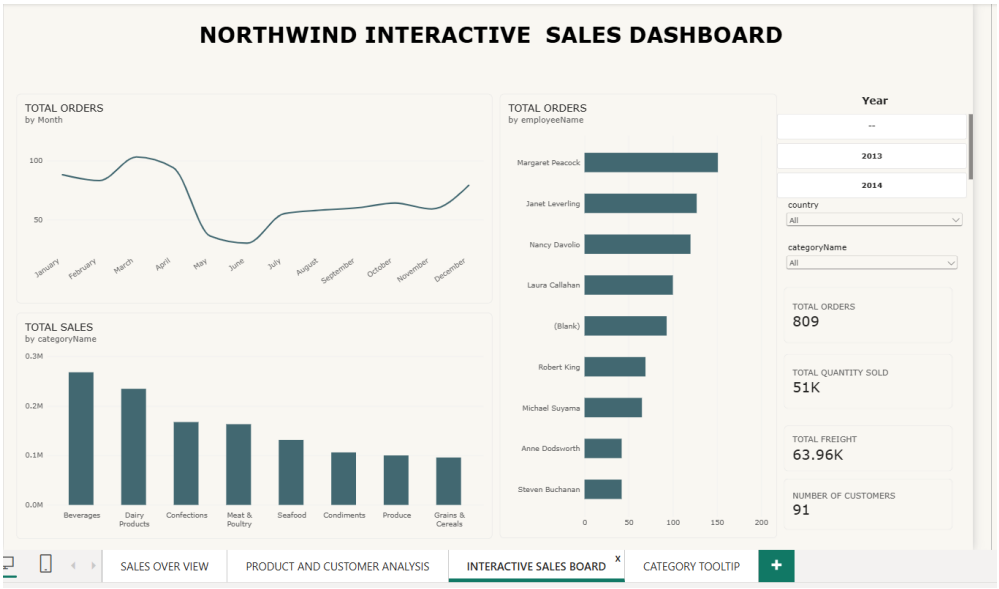


Figure 3: Northwind Interactive Sales Dashboard with year, country and category filters.

9. Importance of the Dashboard

- **Order Monitoring:** KPI cards provide a quick view of total orders and customer count.
- **Monthly Analysis:** The line chart helps me identify changes in order activity throughout the year.
- **Employee Comparison:** The bar chart makes it easier to compare order counts by employee.
- **Category Analysis:** The column chart helps me compare sales across product categories.
- **Customer Analysis:** The country chart shows how customers are distributed geographically.
- **Interactive Filtering:** Year, country and category filters allow me to focus on selected data.

10. Category Tooltip

I created a separate Category Tooltip page containing three KPI cards: Total Sales, Total Quantity Sold and Total Orders. The displayed values are approximately 1.27M, 51K and 809 respectively. When this page is assigned as a tooltip to a related visual, it can provide these summary details when I hover over a data point or category.

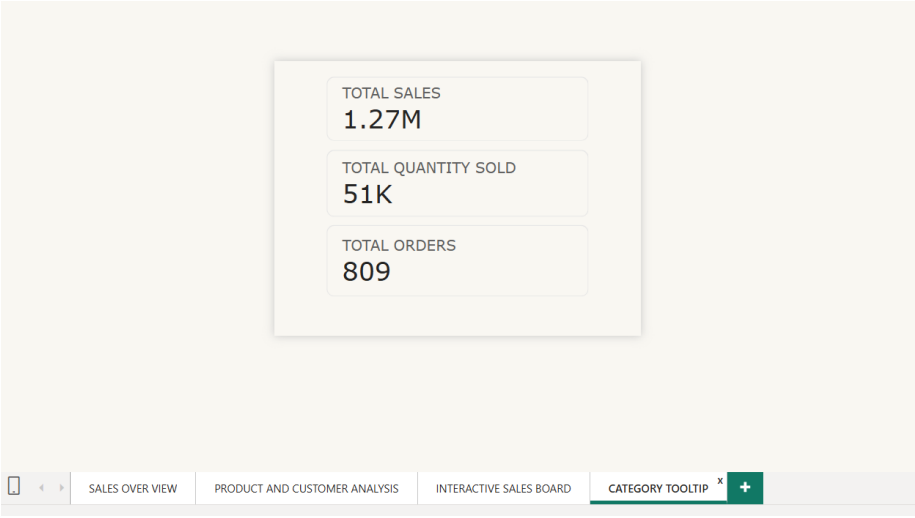


Figure 4: Category Tooltip page with sales, quantity and order summary cards.

11. Conclusion

Through this project, I learned how to use Microsoft Power BI to create an interactive sales dashboard using order, customer, product, employee, category and country information. I used KPI cards to summarize orders, customers, quantity and freight, and created charts to compare monthly order activity, employee orders, category sales and customer counts by country.

The product-level table helped me review sales and quantity for individual products, while the year, country and category filters made it possible to focus on selected parts of the data. I also created a Category Tooltip page to display key sales and order measures in a compact view.

Overall, this project improved my understanding of data analysis, KPI creation, chart selection, interactive filtering and tooltip design using Power BI.

12. References

- Microsoft Power BI Desktop.
- Northwind sales dataset used for the project.
- Microsoft Power BI documentation.